WORKSHEET

Geometric sequences

Rule

In a geometric sequence, the nth term, t_n, can be found by $t_n = ar^{n-1}$, where a is the first term and r is the common ratio.
The common ratio: $r = \frac{t_n}{t_{n-1}}$.

Questions

1 What is the common ratio in each of the following geometric sequences?

a 4, 12, 36, 108, …

b 24, −12, 6, −3, …

c $1200, 800, 533\frac{1}{3}, 355\frac{5}{9}, \ldots$

2 What is the rule for t_n for the following geometric sequences?

a 3, 15, 75, 375, …

b 192, −38.4, 7.68, −1.536, …

c 0.05, 0.3, 1.8, 10.8, …

3 What is the 12th term of the sequence: $\frac{3}{25}, -\frac{3}{5}, 3, -15, \ldots$?

4 What is the seventh term of the sequence: 5, −5, 5, −5, …?

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5 What is the 50th term of the sequence: 80 000, 76 000, 72 200, …?

6 A person begins to train for a 10-km race by running 1 km on the first day and increasing this distance by 10% each day. After how many days will he/she be able to run at least 10 km?

7 A car depreciates by 20% each year. After how many years will a car bought for $28 000 be worth less than $10 000?

8 How many terms are in each of the following geometric sequences?

a 500, 50, 5, …, 0.0005

b $-\frac{3}{4}, 1\frac{1}{2}, -3, \ldots, 6144$

c 20, 21, 22.05, …, 32.577 89

9 A geometric sequence has the terms $t_3 = 180$ and $t_7 = 2\frac{2}{9}$.

a Find the common ratio and the first term of the sequence.

b Find the 10th term of the geometric sequence.

10 Find t_{20} of a geometric sequence if $t_2 = -1$ and $t_5 = 8$.

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Answers

1 **a** 3

b $-\frac{1}{2}$

c $\frac{2}{3}$

2 **a** $t_n = 3 \times 5^{n-1}$

b $192 \times (-0.2)^{n-1}$

c $0.05 \times 6^{n-1}$

3 $-5\,859\,375$

4 5

5 6479.58

6 26

7 6

8 **a** 7

b 14

c 11

9 **a** $r = \frac{1}{3}, a = 1620$

b $\frac{20}{243}$

10 $-262\,144$

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